

Gradients with a numerical check

Project: Image-pattern Network Investigation

1. Your mission

Your learning target

I can estimate one partial derivative while other parameters stay fixed.

I can assemble a gradient in a declared parameter order.

I can compare numerical and formula based gradients with a tolerance.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Slopes and local change. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Parameters and architecture. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Activation functions. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

This week you will practise:

Estimate one partial derivative while other parameters stay fixed

Assemble a gradient in a declared parameter order

Compare numerical and formula based gradients with a tolerance

Retrieve what you already know

Why is the symmetric-slope denominator twice the step?

My starting explanation

This week's end goal

By the end, I can explain gradients with a numerical check using a traced example and a stated limitation.

2. Notice the evidence

Use one example with input $x = 2$ and target $t = 3$. The model predicts $y = wx + b$, and loss $L(w, b) = (2w + b - 3)^2$. At $w = 1$, $b = 0$, prediction is 2, residual is -1 and loss is 1. There are two parameters in order $[w, b]$.

To estimate the partial slope for w , change w only and keep $b = 0$. To estimate the partial slope for b , change b only and keep $w = 1$. Moving both at once measures change along a combined direction; it does not isolate either partial derivative.

Perturbation from $[1, 0]$ · Loss at minus step · Loss at plus step · Symmetric slope

Perturbation from $[1, 0]$: w only with step 0.1

Loss at minus step: 1.44

Loss at plus step: 0.64

Symmetric slope: -4

Perturbation from $[1, 0]$: b only with step 0.1

Loss at minus step: 1.21

Loss at plus step: 0.81

Symmetric slope: -2

The gradient is $[-4, -2]$ in order $[w, b]$. Both entries refer to the same original parameter vector, not to a partially updated model.

List the given facts and one fact that is not supplied.

3. See how it works

```
def loss(w, b):  
    return (2 * w + b - 3) ** 2  
  
def gradient_check(w, b, step=0.001):  
    dw = (loss(w + step, b) - loss(w - step, b)) / (2 * step)  
    db = (loss(w, b + step) - loss(w, b - step)) / (2 * step)  
    residual = 2 * w + b - 3  
    formula = [4 * residual, 2 * residual]  
    numerical = [dw, db]  
    assert all(abs(a - n) < 1e-6  
               for a, n in zip(formula, numerical))  
    return numerical  
  
print([round(g, 6) for g in gradient_check(1, 0)])  
# [-4.0, -2.0]
```

The helper assumes finite numerical inputs and a positive nonzero step. The formula follows the squared-loss rule: local loss change is twice the residual times the prediction sensitivity. Changing w by a small amount changes prediction twice as much here; changing b changes prediction by the same amount.

Exact floating-point equality is not required. Record the step and a tolerance, then try more than one parameter setting. A passing numerical check supports agreement for those tested points; it does not prove all possible inputs.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

Compare a familiar case with a changed case

A w probe has losses 1.44 and 0.64 at 0.9 and 1.1 with b fixed. A learner divides their difference by 0.1. Correct the result.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change: Swap the two gradient components in the program output.

Prediction: The coordinate-wise numerical check exposes the mismatch.

Reason: The w perturbation changes y twice as strongly as the same b perturbation.

5. Words that unlock the idea

Word · Meaning

Word: Partial derivative

Meaning: A local slope with respect to one parameter while others are held fixed

Word: Gradient

Meaning: The ordered vector of partial derivatives of a scalar function

Word: Finite difference

Meaning: An estimate of change computed from nearby function values

Word: Tolerance

Meaning: A stated allowance for numerical disagreement

Use two terms to explain a step in this week’s example.

6. Follow a worked example

Two separate probes at one parameter vector

Parameter being probed · Held fixed · Gradient entry

Parameter being probed: w near 1

Held fixed: b is 0 and x is 2

Gradient entry: First entry -4

Parameter being probed: b near 0

Held fixed: w is 1 and x is 2

Gradient entry: Second entry -2

Parameter being probed: Assemble in order $[w, b]$

Held fixed: Same original model

Gradient entry: Gradient $[-4, -2]$

The first two rows are independent local probes, not successive parameter updates. The final row records both results in the declared order.

At $w = 0$, $b = 1$, find prediction, residual, loss and the gradient using [4 times residual, 2 times residual].

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Gradients with a numerical check

Check each coordinate of a gradient while holding the other parameter fixed.

The facts we will use

Model $y=2w+b$, target 3. At $[w, b]=[1, 0]$, $y = 2$, residual $y-\text{target}=-1$; loss is residual^2 .

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

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Gradients with a numerical check

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Check w direction

Analytic derivative $2 \times \text{residual} \times 2 = -4$.



2. Check b direction

Analytic derivative $2 \times \text{residual} = -2$.



3. Numerical check

With step 0.1: w losses 0.64, 1.44 give -4; b losses 0.81, 1.21 give -2.

CHANGE ONE THING

Swap the two gradient components in the program output.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Check w direction

What happens and why: Analytic derivative $2 \times \text{residual} \times 2 = -4$.

Step: 2. Check b direction

What happens and why: Analytic derivative $2 \times \text{residual} = -2$.

Step: 3. Numerical check

What happens and why: With step 0.1: w losses 0.64, 1.44 give -4; b losses 0.81, 1.21 give -2.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

A gradient lists the slope for each parameter, in a declared order. Changing w by a small amount changes the prediction twice as much as changing b by the same amount. That explains the different derivative magnitudes. Perturb one coordinate while holding the other fixed, then compare the numerical slope with its analytic counterpart.

What this example does not establish

The exact agreement occurs for this quadratic toy case; finite differences in larger models involve rounding and step-size limits.

7. Find and repair a mistake

ReLU is not differentiable exactly at zero. A symmetric difference across that corner can average two different behaviours. When checking a network gradient, use points whose relevant scores are safely away from zero, and check that the small perturbations do not cross a corner. Document any derivative convention used at zero.

At the centre of the simple scalar ReLU function, $[\text{ReLU}(0.1) - \text{ReLU}(-0.1)] / 0.2$ equals 0.5. This symmetric number does not make the function differentiable at zero. Also avoid assuming an ever-smaller step is always better: subtraction and rounding can spoil numerical differences.

Locate the earliest unsupported step

Starting at $[w, b] = [1, 0]$, both coordinates move by plus or minus 0.1 together. Compute the symmetric slope and explain why it is not either partial derivative.

State one repair and the checks you would rerun.

8. Practise together

For the lesson model at $w = 1$ and $b = 1$, calculate prediction, residual, loss and gradient.

At $w = 0$, $b = 1$, find prediction, residual, loss and the gradient using [4 times residual, 2 times residual].

Explain your prediction to a partner before comparing results.

Practise with less help: Catch a denominator error

Use workbook W2 for the connected example “Gradients with a numerical check”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Gradients with a numerical check. Proposed explanation: “The gradient is one scalar”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why should the numerical checks perturb one coordinate at a time? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

Predict, investigate and preserve the evidence

A w probe has losses 1.44 and 0.64 at 0.9 and 1.1 with b fixed. A learner divides their difference by 0.1. Correct the result.

```
def loss(w, b):  
    return (2 * w + b - 3) ** 2  
  
def gradient_check(w, b, step=0.001):  
    dw = (loss(w + step, b) - loss(w - step, b)) / (2 * step)  
    db = (loss(w, b + step) - loss(w, b - step)) / (2 * step)  
    residual = 2 * w + b - 3  
    formula = [4 * residual, 2 * residual]  
    numerical = [dw, db]  
    assert all(abs(a - n) < 1e-6  
               for a, n in zip(formula, numerical))  
    return numerical
```

```
print([round(g, 6) for g in gradient_check(1, 0)])  
  
# [-4.0, -2.0]
```

Record · Your evidence

Record: Given input and fixed rule

Your evidence: _____

Record: Prediction before checking

Your evidence: _____

Record: Actual result, or “not run”

Your evidence: _____

Record: First mismatch and one explanation

Your evidence: _____

Record: Boundary or missing input and expected response

Your evidence: _____

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

Explain → apply → challenge

For the lesson model at $w = 1$ and $b = 1$, calculate prediction, residual, loss and gradient.

Starting at $[w, b] = [1, 0]$, both coordinates move by plus or minus 0.1 together. Compute the symmetric slope and explain why it is not either partial derivative.

Compute the symmetric difference of ReLU at zero with step 0.1. Explain why this does not prove a derivative exists there.

Explain which evidence supports your conclusion and which claim remains unproved.

11. Connect your project

Add a labelled gradients with a numerical check evidence sheet to Image-pattern Network Investigation. Show how you can estimate one partial derivative while other parameters stay fixed. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence · Record

Project evidence: Component and version

Record: _____

Project evidence: Inputs permitted at prediction time

Record: _____

Project evidence: Trace and comparison with earlier work

Record: _____

Project evidence: Failure, limit and human review

Record: _____

Evidence for your project

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when

checking independent understanding.

12. Show what you know

The gradient describes how the stated loss changes near the current parameters. For a differentiable function, moving a sufficiently small distance opposite a nonzero gradient decreases the function locally. The gradient alone does not choose a safe finite step, establish a global minimum or demonstrate generalisation.

A numerical checker moves w and b together, gets one slope and copies it into both gradient entries. What is wrong with this check?

Next week combines a gradient with an explicit learning rate to update parameters together.

Compute the symmetric difference of ReLU at zero with step 0.1. Explain why this does not prove a derivative exists there.

My limited claim and a useful next check

13. Study a complete case

Complete evidence record

Two separate probes at one parameter vector

Parameter being probed · Held fixed · Gradient entry

Parameter being probed: w near 1

Held fixed: b is 0 and x is 2

Gradient entry: First entry -4

Parameter being probed: b near 0

Held fixed: w is 1 and x is 2

Gradient entry: Second entry -2

Parameter being probed: Assemble in order [w, b]

Held fixed: Same original model

Gradient entry: Gradient [-4, -2]

The first two rows are independent local probes, not successive parameter updates. The final row records both results in the declared order.

Follow the input through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

14. Try a new case independently

Independent case: Check a new example independently

Use $x = 3$, target 2 and $y = wx + b$ at $w = 0$, $b = 1$. With step 0.1, calculate both numerical partials. Compare with [2 times residual times x, 2 times residual].

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Use $x = 3$, target 2 and $y = wx + b$ at $w = 0$, $b = 1$. With step 0.1, calculate both numerical partials. Compare with [2 times residual times x , 2 times residual].

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.